

MES BUILD PLAN — EV Motor Assembly | Path 3 (Custom + Tally)

Greenfield · 500→2000 motors/month · Team-built with Claude

Production: Aug start
500 → 2000 /mo

Stack: Postgres +
FastAPI + React

PLC capture at
critical stations

Tally stays as ERP
(no replacement)

PHASE 1 — MVP for August Go-Live (Build NOW)

Master Data
Parts, BOMs, Workstations,
Operators, Suppliers, Customers

Incoming GRN
Supplier lot capture,
Quarantine → QC release

Production Order
Create PO, auto-generate
serial numbers

Station Tracking
Scan serial + badge,
record pass/fail + cycle time

Component Consume
Scan component lots
at each station

PLC Data Capture
OPC-UA / Modbus service
stamps readings to serial

End-of-Line Test
Capture all test params
vs limits → pass/fail

Dispatch + Trace
Scan to pack, dispatch note,
full genealogy query

PHASE 2 — Months 2-4 After Launch

Defect Pareto
Trends by machine,
operator, shift, part

Operator Reports
Productivity, cycle times,
rejection rates

WIP Visibility
Live dashboard of
motors at each station

Rework Workflow
NCR, disposition,
root cause logging

Tally Auto-Post
Production receipts,
scrap, FG entries via XML

PHASE 3 — As You Scale Toward 2000/month

OEE Dashboard
Availability, performance,
quality per station

SPC on PLC Data
Control charts on torque,
press force, leak rates

Calibration Track
Test bench & gauge
calibration schedules

ANDON Alerts
Real-time alerts on
failures, downtime

FMEA Linkage
Connect defects back
to FMEA causes

TECHNICAL ARCHITECTURE

SHOP FLOOR
Barcode scanners
Touch tablets
Label printers
(Zebra/TSC)

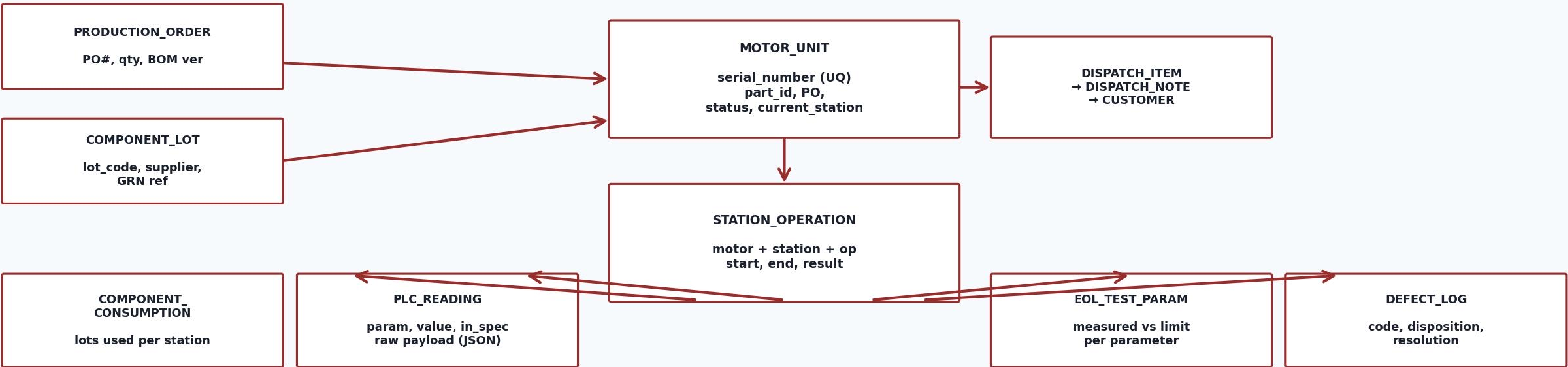
PLC LAYER
OPC-UA / Modbus
Python poller svc
Tags → DB

APP LAYER
FastAPI backend
React frontend
Local server +
offline cache

DATA LAYER
PostgreSQL
Audit log (IATF)
Nightly cloud
backup

INTEGRATION
Tally XML/ODBC
Customer ASN
Label pipelines

CORE DATA MODEL — Traceability Spine



CRITICAL SUCCESS FACTORS — Things That Will Sink You

Scope Discipline
Build MVP only.
Resist feature creep.

Scan Discipline
Operators MUST scan.
Process > Software.

Offline Mode
Network WILL fail.
Local cache + sync.

Tested Backups
Untested backup =
No backup. Drill it.

Audit Trail
Immutable logs.
IATF demands it.

Start narrow · Get one motor through end-to-end before scaling · Use Claude for code, your team for judgment